Immune Landscape and Immune Escape States in Lung Squamous Cell Carcinoma Revealed by Transcriptomic Immune Signatures

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Abstract

Tumor immune microenvironments display substantial heterogeneity that influences tumor progression and response to immunotherapy. However, the organization of immune states and their relationship to immune escape mechanisms in lung squamous cell carcinoma (LUSC) remains incompletely understood.

Here, we performed a computational analysis of tumor immune landscapes using RNA-seq–derived immune gene signatures representing tumor inflammation (TIS), cytotoxic T-cell activity, tertiary lymphoid structures (TLS), and T-cell exhaustion. Dimensionality reduction analysis revealed distinct immune phenotypes across tumors, including inflamed, cold, TLS-rich, and exhausted states.

Machine learning analysis using a Random Forest classifier identified the tumor inflammation signature (TIS) as the dominant driver of immune subtype classification.

Immune activation strongly correlated with checkpoint signaling, indicating a potential negative feedback regulatory mechanism limiting anti-tumor immunity.

An immune state model based on cytotoxicity and exhaustion signatures suggested a potential progression from immune-cold tumors through cytotoxic activation toward dysfunctional exhausted states characterized by high checkpoint signaling.

These findings suggest that chronic immune stimulation in LUSC tumors may promote progressive T-cell exhaustion and contribute to immune escape.

Introduction

Immune checkpoint blockade therapies have transformed the treatment landscape of multiple cancers, including lung cancer. However, clinical responses remain highly heterogeneous, highlighting the importance of understanding tumor immune microenvironment organization.

Lung squamous cell carcinoma (LUSC), a major subtype of non-small cell lung cancer, exhibits diverse immune phenotypes that influence tumor progression and therapeutic response. Recent studies have shown that immune infiltration, cytotoxic activity, and checkpoint signaling play key roles in shaping tumor immune states.

Gene expression–based immune signatures provide a powerful approach to quantify immune activity within tumors and characterize immune heterogeneity across patient cohorts. In particular, tumor inflammation signature (TIS), cytotoxic T-cell activity, tertiary lymphoid structure (TLS) signaling, and T-cell exhaustion programs capture complementary aspects of anti-tumor immunity.

In this study, we used computational transcriptomic analysis to investigate immune landscape organization in LUSC tumors. By integrating immune gene signatures, dimensionality reduction, and machine learning approaches, we sought to identify immune phenotypes and explore potential trajectories leading to immune dysfunction and escape.

Results

Immune landscape of LUSC tumors reveals distinct immune phenotypes

The final dataset comprised 32 LUSC tumors, which were analyzed using immune gene signature scoring and machine learning approaches.

To characterize the tumor immune microenvironment, we quantified four immune gene signatures representing key immune processes: tumor inflammation signature (TIS), cytotoxic T-cell activity, T-cell exhaustion, and tertiary lymphoid structure (TLS) signaling.

Dimensionality reduction of signature scores using UMAP revealed substantial heterogeneity in the immune landscape across tumor samples (Figure 1). Tumors formed distinct clusters in the immune landscape representation, indicating variability in immune activation and immunoregulatory pathways across the cohort.

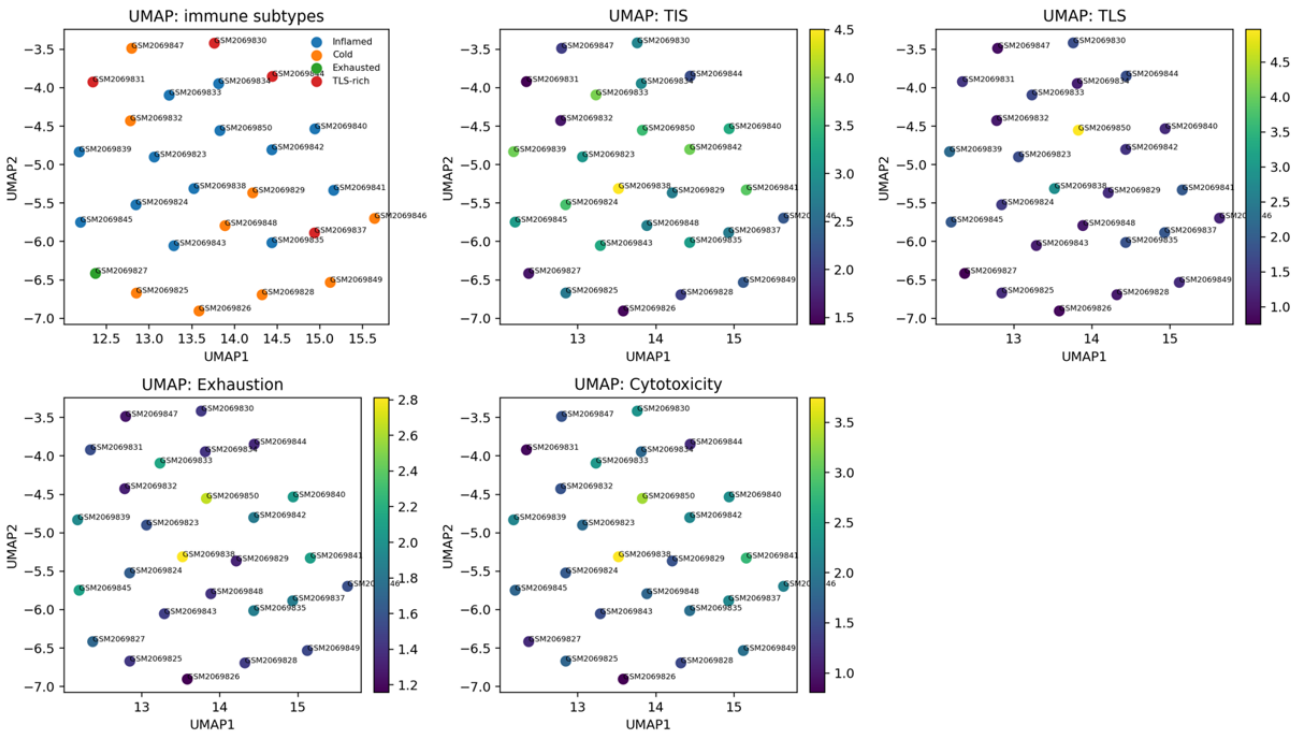

Figure 1. Immune landscape of LUSC tumors based on immune gene signatures.

Based on signature profiles, tumors were classified into four immune subtypes: Inflamed, Cold, TLS-rich, and Exhausted. Inflamed tumors exhibited the highest levels of immune activation and cytotoxic signaling, whereas Cold tumors displayed minimal immune infiltration and low immune activity. TLS-rich tumors showed elevated TLS-associated signatures, while Exhausted tumors were characterized by increased expression of exhaustion-related markers.

Machine learning identifies immune activation as the dominant determinant of immune subtype

To determine which immune features best explain subtype variability, we trained a Random Forest classifier using the four immune signatures. The model achieved a cross-validated accuracy of approximately 0.74, indicating that immune signatures reliably distinguish tumor immune phenotypes.

Feature importance analysis revealed that TIS was the strongest driver of subtype classification, followed by TLS and cytotoxicity signatures.

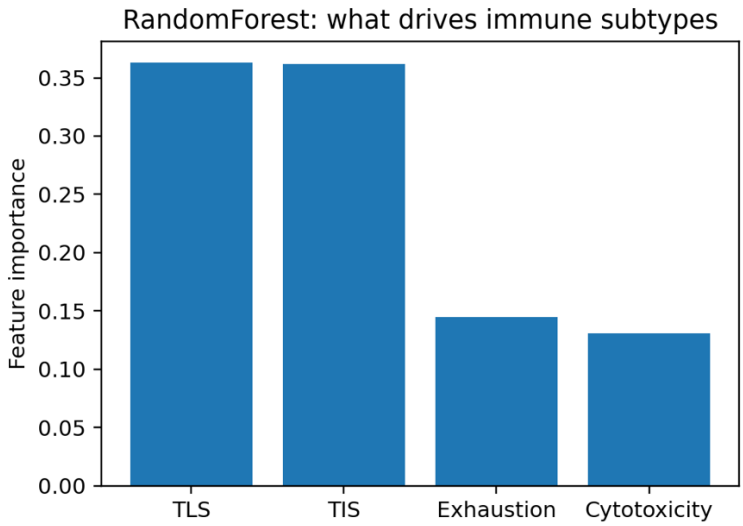

Figure 2. Machine learning analysis identifying immune activation as the major determinant of immune subtype classification.

Feature importance from the Random Forest model highlights tumor inflammation signature (TIS) as the dominant driver.

The exhaustion signature contributed less strongly but still provided discriminatory information between inflamed and exhausted phenotypes.

Together, these findings suggest that global immune activation, captured by the tumor inflammation signature, represents the primary axis structuring immune heterogeneity in LUSC tumors.

Immune activation strongly correlates with checkpoint signaling

To investigate the relationship between immune activation and inhibitory immune pathways, we calculated a checkpoint signaling score based on the expression of eight key immune checkpoint genes:

PDCD1, CD274, CTLA4, LAG3, TIGIT, HAVCR2, ENTPD1, and TOX.

Across tumors, immune activation showed a strong positive correlation with checkpoint signaling (Pearson $r = 0.84$, Spearman $\rho = 0.79$, $R^2 = 0.70$; Figure 3). Tumors with higher immune activation consistently exhibited elevated expression of checkpoint molecules.

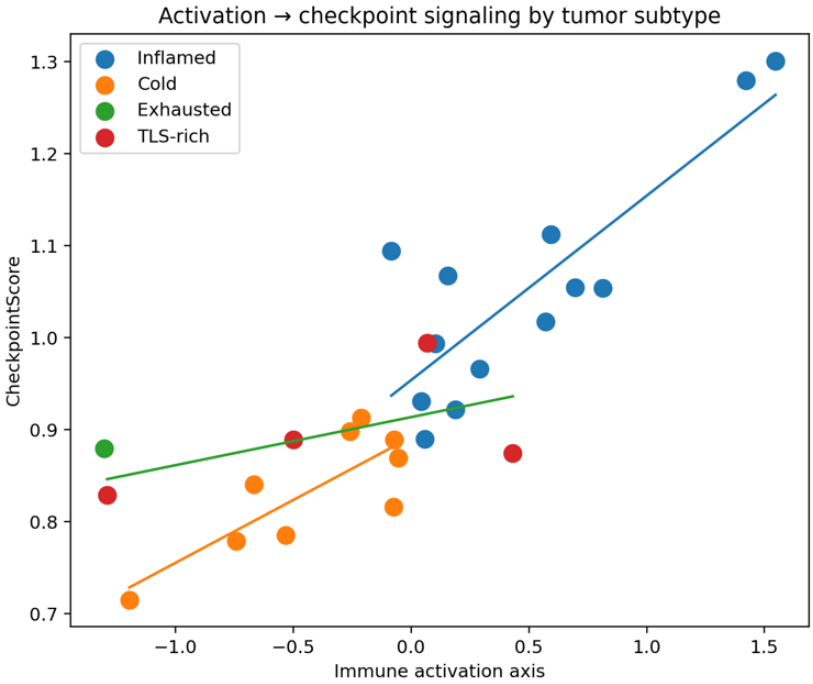

Figure 3. Scatter plot illustrating the relationship between immune activation and checkpoint signaling across tumors.

This pattern suggests the presence of a negative feedback regulatory mechanism, in which increased T-cell activation induces the expression of inhibitory immune checkpoints that limit anti-tumor immune responses.

Immune state diagram reveals functional and dysfunctional immune states

To further characterize tumor immune behavior, we constructed an immune state diagram based on cytotoxic activity and exhaustion signatures. Tumors were categorized into four immune states:

- Cold (low cytotoxicity, low exhaustion)
- Functional (high cytotoxicity, low exhaustion)
- Suppressed (low cytotoxicity, high exhaustion)
- Exhausted (high cytotoxicity, high exhaustion)

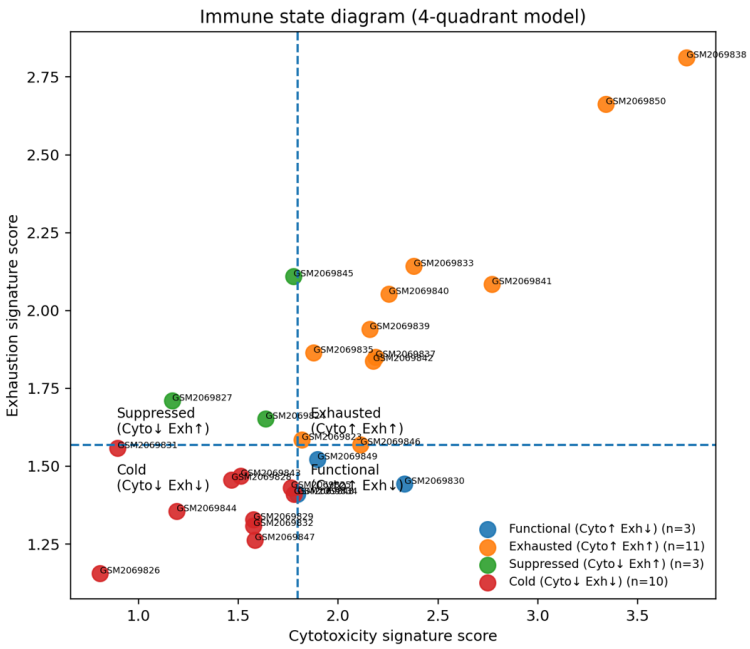

Figure 4. Immune state diagram illustrating functional and dysfunctional immune states defined by cytotoxic activity and exhaustion signatures.

Two dominant immune states were observed: exhausted tumors (n = 11) and cold tumors (n = 10), whereas functional and suppressed states were less frequent.

Importantly, immune states were strongly associated with tumor immune subtypes. Inflamed tumors were predominantly found in the exhausted state, whereas Cold subtype tumors clustered in the cold immune state. This observation suggests that immune infiltration in inflamed tumors frequently progresses toward T-cell dysfunction.

Checkpoint signaling differs significantly across immune states

Checkpoint signaling levels varied significantly between immune states (Kruskal–Wallis test, $H = 18.71$, $p = 3.1 \times 10^{-4}$). Pairwise comparisons showed that exhausted tumors exhibited significantly higher checkpoint signaling than cold tumors ($p = 1.2 \times 10^{-4}$) and functional tumors ($p = 0.022$).

These findings demonstrate that checkpoint expression is strongly associated with T-cell exhaustion rather than cytotoxic activation alone.

Immune state analysis suggests progression toward immune exhaustion

To integrate these observations, we constructed a conceptual immune state model linking cytotoxic activation and exhaustion signatures. The model suggests a potential progression from immune-cold tumors lacking immune infiltration through functionally active cytotoxic states toward exhausted immune states characterized by high checkpoint signaling (Figure 5).

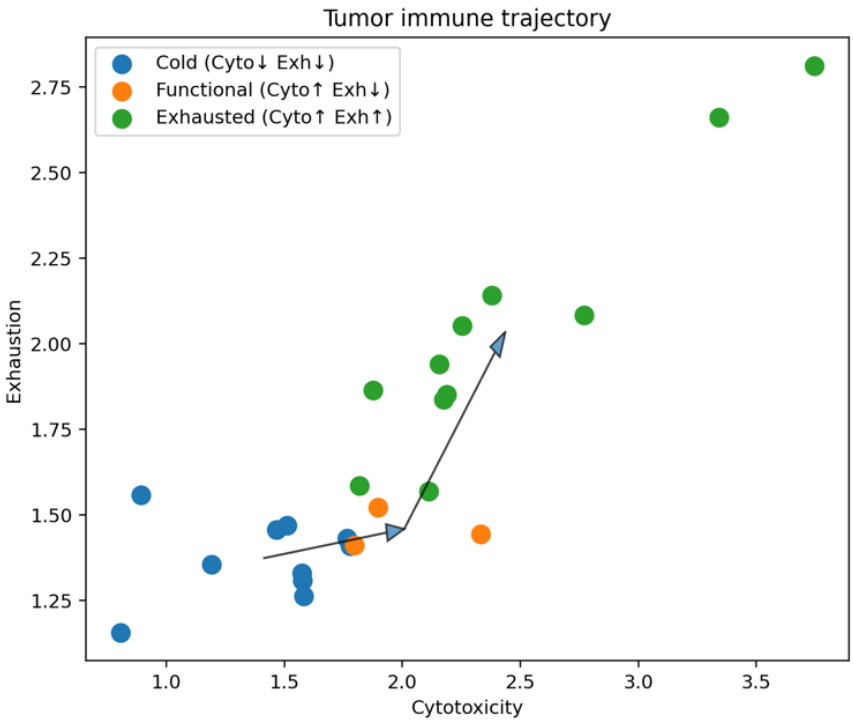

Figure 5. Conceptual immune state model linking cytotoxic activation to progressive T-cell exhaustion.

This model supports the hypothesis that chronic immune stimulation in inflamed tumors drives progressive T-cell exhaustion and activation of immune checkpoint pathways, ultimately contributing to tumor immune escape.

Methods

Dataset and transcriptomic data processing

Transcriptomic data for lung squamous cell carcinoma (LUSC) were obtained from The Cancer Genome Atlas (TCGA-LUSC cohort). The final dataset included 32 tumor samples with available gene expression profiles. Only samples with complete expression data required for immune signature scoring were retained for downstream analysis.

Normalized gene expression matrices were used for all downstream analyses. Genes with very low or undetectable expression across samples were excluded to reduce noise and improve statistical robustness.

All analyses were conducted in Python (version 3.x) using the Jupyter Notebook environment. Data processing and statistical analyses were performed using standard scientific computing libraries including pandas, numpy, and scipy.

Dimensionality reduction was performed using UMAP implemented in the Python umap-learn package. Data visualization and exploratory analyses were carried out using matplotlib, seaborn, and plotly.

Machine learning analyses were performed using the scikit-learn framework, and model interpretability was evaluated using SHAP (SHapley Additive exPlanations).

Immune hot and cold tumor classification

Tumors were stratified into immune-hot and immune-cold groups based on the expression patterns of genes associated with immune cell infiltration and immune activity within the tumor microenvironment. Immune-hot tumors were characterized by increased expression of genes associated with adaptive and innate immune responses, including markers of T lymphocytes, B cells, natural killer cells, and antigen presentation pathways. In contrast, immune-cold tumors displayed lower expression of immune-related genes and reduced signatures of immune infiltration.

This classification enabled the comparison of transcriptional profiles between tumors with active immune microenvironments and those with limited immune engagement.

Differential gene expression analysis

Differential gene expression between immune-hot and immune-cold tumor groups was evaluated using Welch's t-test, which accounts for unequal variance between groups. Gene expression differences were quantified using fold change values and statistical significance.

Genes were ranked according to their differential expression, and the results were compiled into a differential expression table. Genes showing the strongest positive fold changes were considered enriched in immune-hot tumors, whereas genes with negative fold changes were associated with immune-cold tumors.

Immune signature analysis

To characterize the tumor immune microenvironment, gene signatures representing major immune cell populations and immune regulatory pathways were evaluated. These included markers associated with:

- T lymphocytes
- B lymphocytes
- natural killer (NK) cells
- macrophages
- immune checkpoint molecules

Signature scores were calculated based on aggregated expression levels of genes belonging to each immune-related category. These scores were used to compare immune activity between immune-hot and immune-cold tumors and to identify pathways associated with immune escape mechanisms.

Machine learning and model interpretability

Machine learning analyses were performed using the scikit-learn framework to explore transcriptional patterns associated with immune phenotypes. Feature importance and model interpretability were assessed using SHAP (SHapley Additive exPlanations), which quantifies the contribution of individual genes to the predictive model.

SHAP analysis enabled the identification of genes with the strongest influence on the classification of immune-hot versus immune-cold tumors, providing insights into potential biomarkers associated with immune escape.

Data visualization

Gene expression patterns and immune signatures were visualized using multiple graphical approaches. Bar plots were used to display the most differentially expressed genes between immune phenotypes. Additional visualizations included gene expression summaries and immune signature plots to illustrate differences in immune activity across tumor groups.

All figures were generated using matplotlib, seaborn, and plotly within the Python environment.

Discussion

Tumor immune microenvironments exhibit substantial heterogeneity that strongly influences tumor progression and response to immunotherapy. In this study, we used transcriptomic immune signatures to characterize the immune landscape of lung squamous cell carcinoma (LUSC) tumors and to investigate relationships between immune activation, checkpoint signaling, and immune dysfunction.

Our analysis revealed distinct immune phenotypes across tumors, including inflamed, cold, TLS-rich, and exhausted states. These phenotypes reflect major organizational axes of the tumor immune microenvironment and highlight the diversity of immune responses present within the LUSC tumor microenvironment. Inflamed tumors displayed strong immune activation and cytotoxic signaling, whereas cold tumors exhibited minimal immune infiltration and immune activity. These findings are consistent with previous studies demonstrating that tumors can vary widely in their degree of immune infiltration and immunogenicity.

Machine learning analysis using a Random Forest classifier identified the tumor inflammation signature (TIS) as the dominant determinant of immune subtype classification. This observation suggests that global immune activation represents a central organizing principle of tumor immune heterogeneity. While cytotoxicity and TLS signatures also contributed to subtype discrimination, TIS captured the largest proportion of variance between immune phenotypes, indicating that coordinated activation of inflammatory pathways plays a critical role in shaping tumor immune landscapes.

A key finding of this study is the strong relationship between immune activation and checkpoint signaling. Tumors with high immune activation consistently exhibited elevated expression of immune checkpoint molecules, including PDCD1, CD274, CTLA4, LAG3, TIGIT, HAVCR2, ENTPD1, and TOX. This pattern suggests the presence of a negative feedback regulatory mechanism in which increased T-cell activation induces inhibitory checkpoint pathways that limit immune responses. Such feedback mechanisms are well recognized in chronic immune stimulation and may contribute to immune escape in tumors with sustained immune infiltration.

To further investigate immune dysfunction, we constructed an immune state model based on cytotoxic activity and exhaustion signatures. This framework identified four immune states—cold, functional, suppressed, and exhausted—representing different functional configurations of anti-tumor immunity. Importantly, exhausted tumors exhibited significantly higher checkpoint signaling than other immune states, indicating that checkpoint expression is closely associated with T-cell dysfunction rather than cytotoxic activity alone.

Integrating these observations, we propose a conceptual immune state model linking immune activation to progressive T-cell exhaustion. In this model, tumors may transition from immune-cold states lacking immune infiltration toward cytotoxic immune activation. However, sustained immune stimulation may subsequently drive expression of inhibitory checkpoint molecules and exhaustion-associated transcriptional programs, ultimately leading to dysfunctional immune states and immune escape. This trajectory framework provides a potential explanation for why tumors with strong immune infiltration can still evade immune destruction.

The findings presented here have important implications for immunotherapy. Immune checkpoint blockade therapies target inhibitory signaling pathways such as PD-1 and CTLA-4 in order to restore anti-tumor immune responses. Our results suggest that tumors with strong immune activation but elevated checkpoint signaling may represent tumors that are primed for checkpoint blockade therapy, whereas immune-cold tumors lacking immune infiltration may require alternative therapeutic strategies aimed at inducing immune activation.

Several limitations should be considered when interpreting these results. First, the analysis was based on transcriptomic immune signatures rather than direct measurements of immune cell populations. Although gene expression signatures provide valuable insights into immune programs, they cannot fully capture spatial organization and functional states of immune cells within the tumor microenvironment. Second, the proposed immune trajectory model represents a conceptual framework derived from cross-sectional transcriptomic data rather than direct temporal observations of tumor evolution. Another limitation of this study is the relatively small sample size (n = 32), which may limit the generalizability of the findings.

Future studies integrating single-cell transcriptomics, spatial transcriptomics, and longitudinal tumor profiling will be important to further elucidate the dynamic processes governing immune activation and exhaustion in tumors.

In conclusion, this study provides a computational framework for characterizing tumor immune landscapes and identifying immune states associated with immune escape in lung squamous cell carcinoma. Our results highlight the central role of immune activation in structuring tumor immune heterogeneity and suggest that chronic immune stimulation may drive progressive T-cell exhaustion through checkpoint signaling pathways. Understanding these processes may help guide the development of more effective immunotherapeutic strategies.

Data Availability

All data analyzed in this study are publicly available. Transcriptomic data for lung squamous cell carcinoma (LUSC) were obtained from The Cancer Genome Atlas (TCGA). A subset of

samples with complete immune signature information was used for the analyses described in this study.

Processed data used for the analysis, immune gene signature lists, and derived results are available in the project repository.

Code Availability

Processed data used for the analysis, immune gene signature lists, and derived results are available in the project repository:

https://github.com/ag48665/lusc-immune-escape-analysis

Figure Legends

Figure 1. Immune landscape of LUSC tumors based on immune gene signatures. UMAP projection of immune signature scores (tumor inflammation signature, cytotoxicity, TLS, and exhaustion) reveals heterogeneous immune phenotypes across lung squamous cell carcinoma tumors. Tumors form distinct clusters corresponding to inflamed, cold, TLS-rich, and exhausted immune states.

Figure 2. Machine learning analysis identifying immune activation as the major determinant of immune subtype classification.

Random Forest feature importance analysis shows that the tumor inflammation signature (TIS) is the dominant predictor of immune subtype classification, followed by TLS and cytotoxicity signatures.

Figure 3. Correlation between immune activation and checkpoint signaling. Scatter plot illustrating the relationship between immune activation and checkpoint signaling score across tumors. A strong positive correlation indicates that tumors with higher immune activation exhibit increased expression of inhibitory checkpoint molecules.

Figure 4. Immune state diagram illustrating functional and dysfunctional immune states.

Tumors are classified into four immune states based on cytotoxic activity and exhaustion signatures: cold, functional, suppressed, and exhausted. The diagram highlights differences in immune activation and T-cell dysfunction across states.

Figure 5. Conceptual immune state model linking cytotoxic activation to progressive T-cell exhaustion.

Proposed model illustrating a potential progression from immune-cold tumors through cytotoxic immune activation toward dysfunctional exhausted immune states characterized by elevated checkpoint signaling and immune escape.

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